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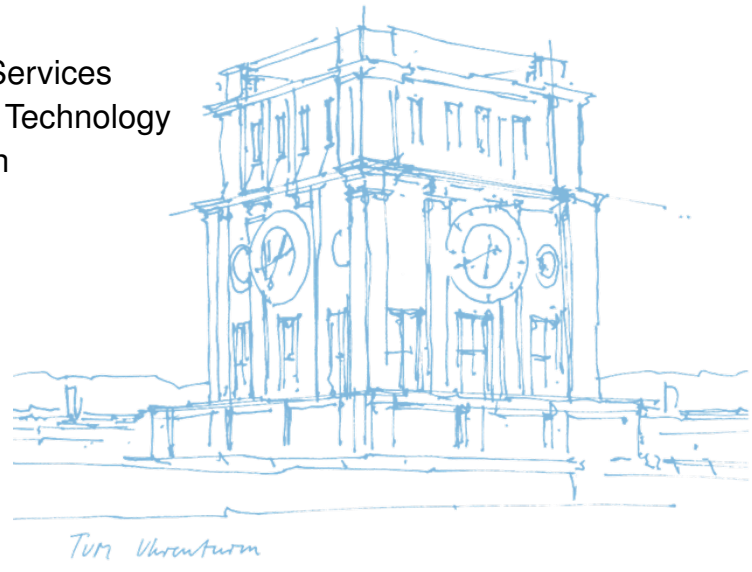
QUIC Improvements Using XDP

Intermediate talk for the Master's Thesis by

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advised by Johannes Späth, Marcel Kempf

Wednesday 23rd April, 2025

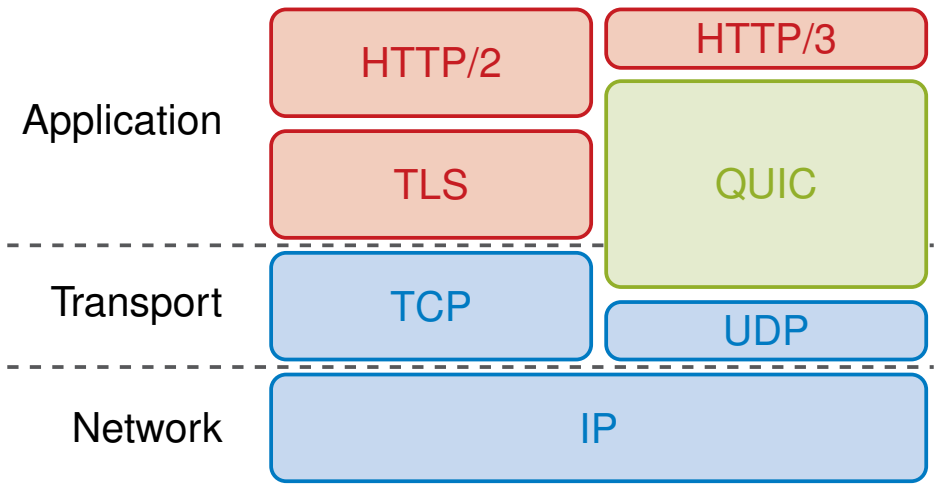
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Motivation

- HTTP/3 commands ~20% of internet traffic worldwide[1]
- Uses QUIC as its L4 protocol
- QUIC throughput is not able to match TCP with increasing bandwidth[2]
- Packet I/O is QUIC's largest bottleneck in terms of its maximum achievable throughput[3]

XDP offers a remedy



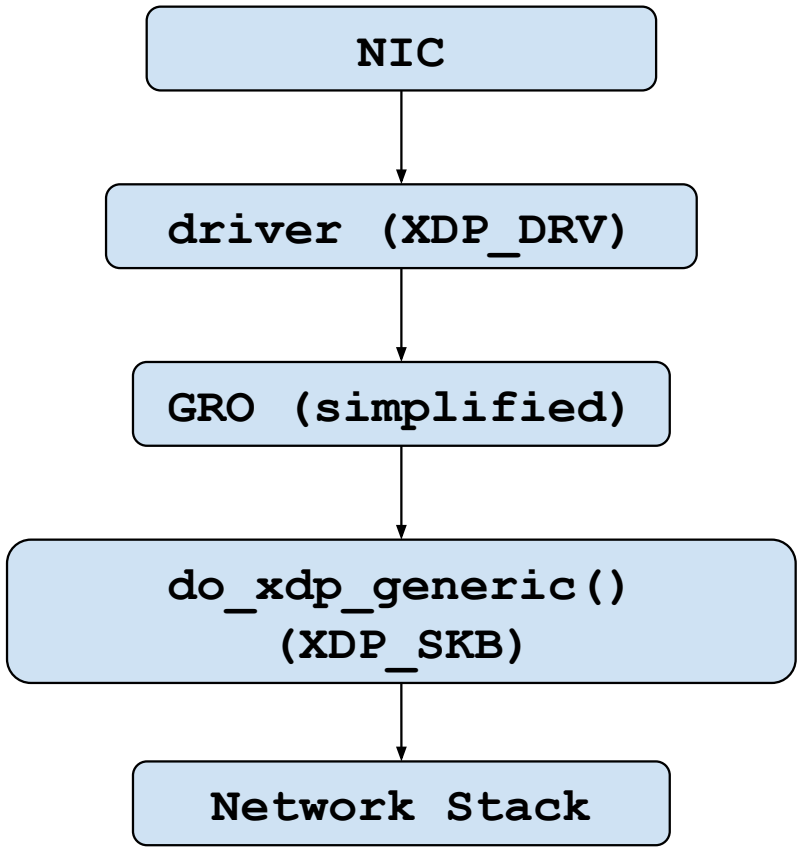
[1] https://radar.cloudflare.com/year-in-review/2024#http-versions

[2] Zhang et al., Quic is not quick enough over fast internet, WWW '24, pp. 2713–2722

[3] Jaeger et al., Quic on the highway, June 2023, pp. 1–9.

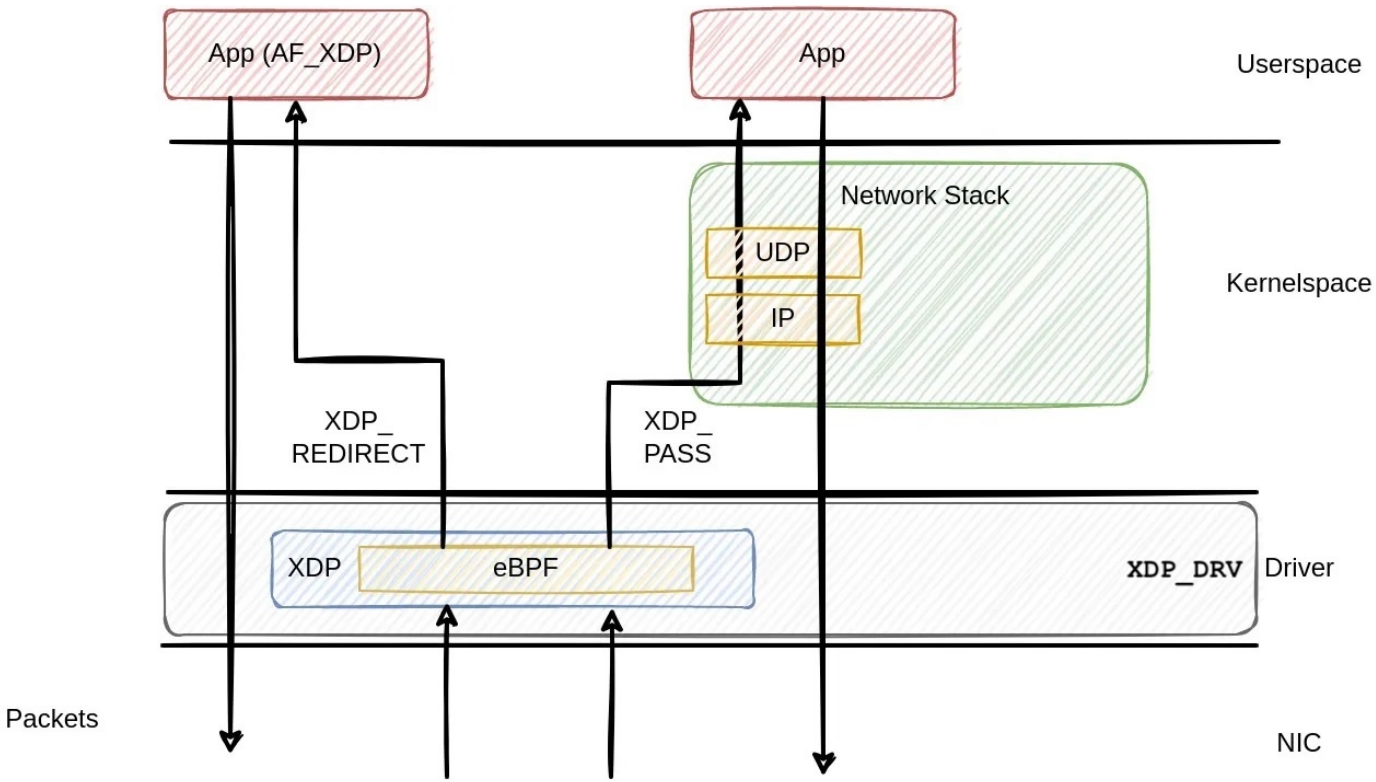
XDP Primer

- XDP is a native Linux hook point for eBPF bytecode to run in
- Decide on an action for each packet: `XDP_DROP`, `XDP_PASS`, `XDP_REDIRECT`, `XDP_ABORTED`, `XDP_TX`



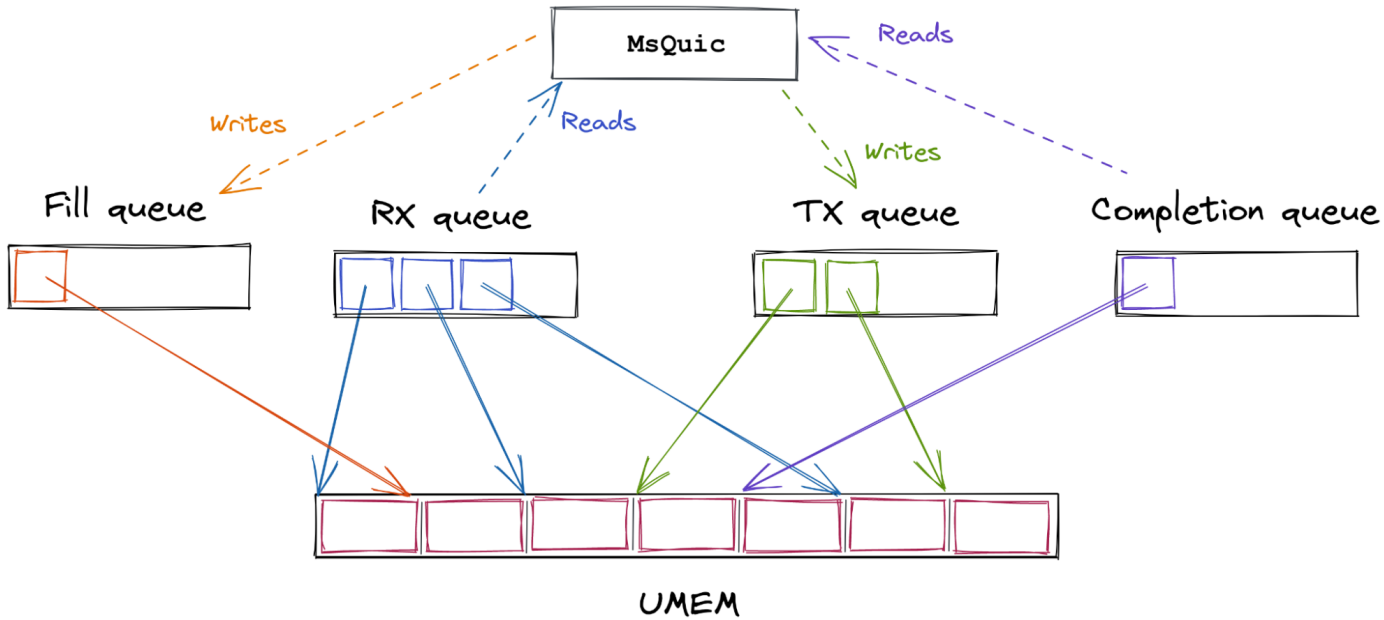
XDP Primer

- AF_XDP is an address family in Linux similar to AF_INET or AF_INET6
- Works with XDP to bypass the network stack
- Helps avoid costly per packet processing



AF_XDP Primer

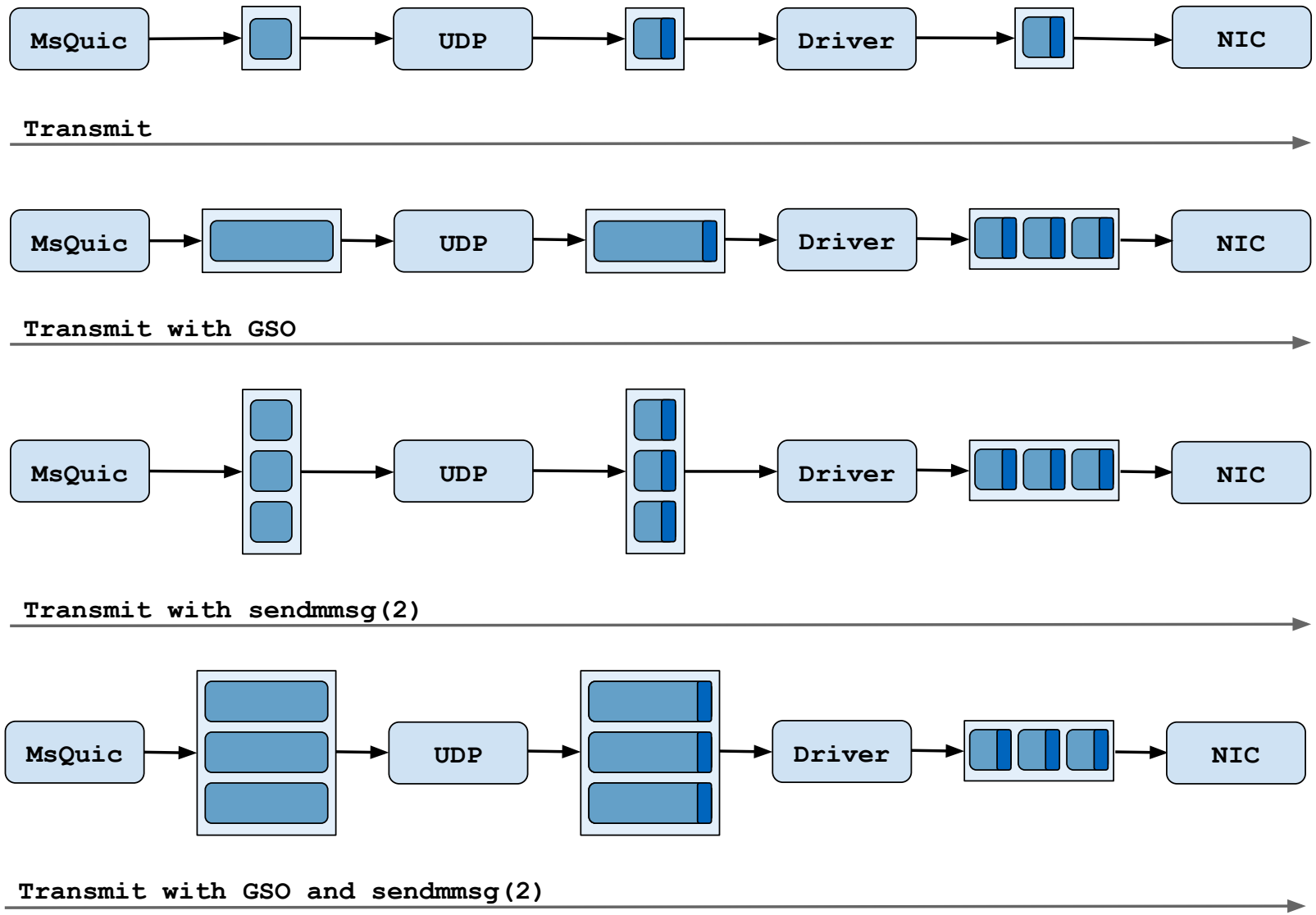
- AF_XDP uses UMEM with tx, rx, completion, and fill rings
- Each ring is single producer, single consumer
- Avoids locking contention
- Need a single system call to trigger multi-packet sends



Additional Optimizations

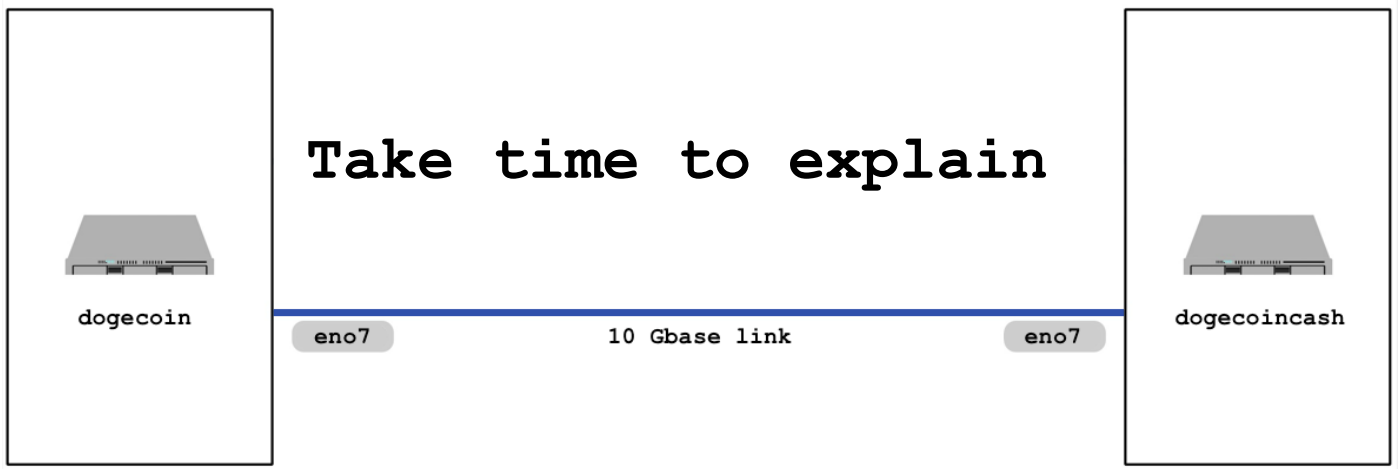
GSO and sendmmsg(2)

Split each part into a different slide



Approach

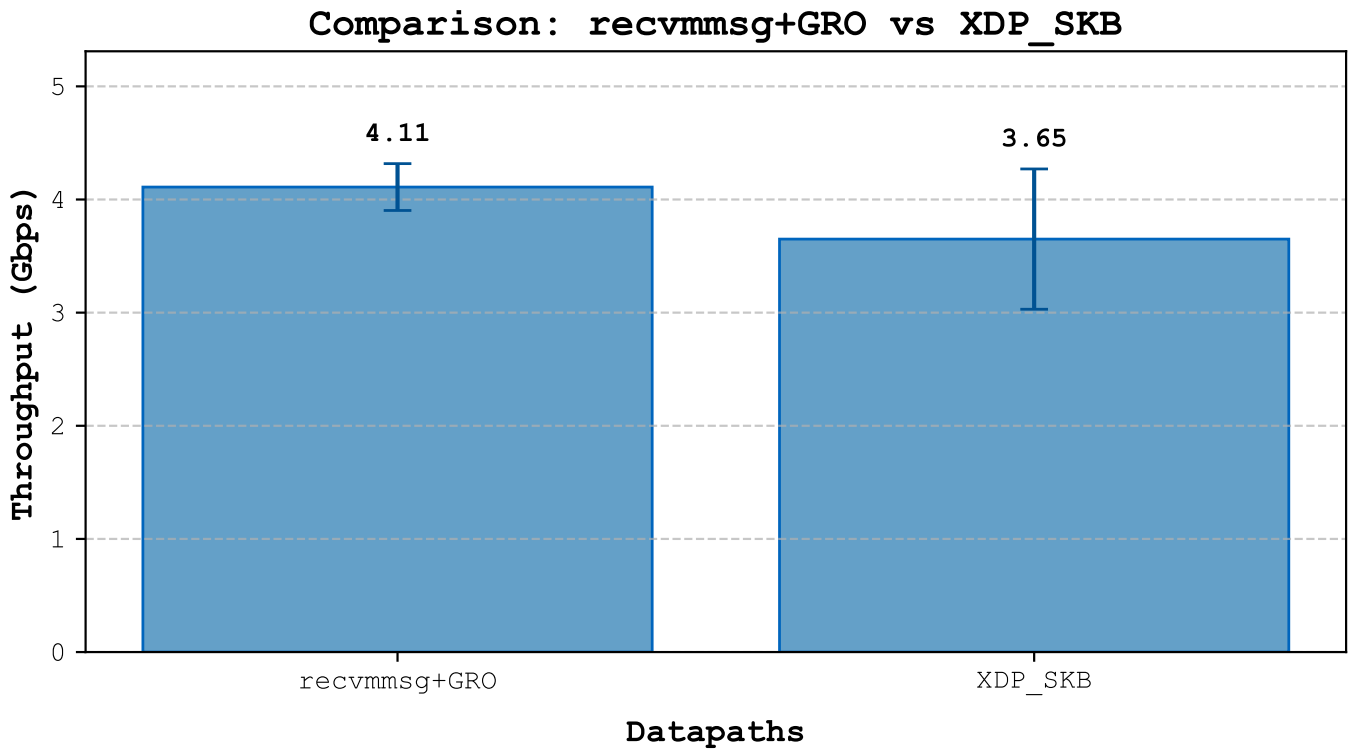
- Patch MsQuic[4] to support AF_XDP
- Run throughput tests in a client-server topology
- 20 iterations with 8GB file size



[4] https://github.com/microsoft/msquic

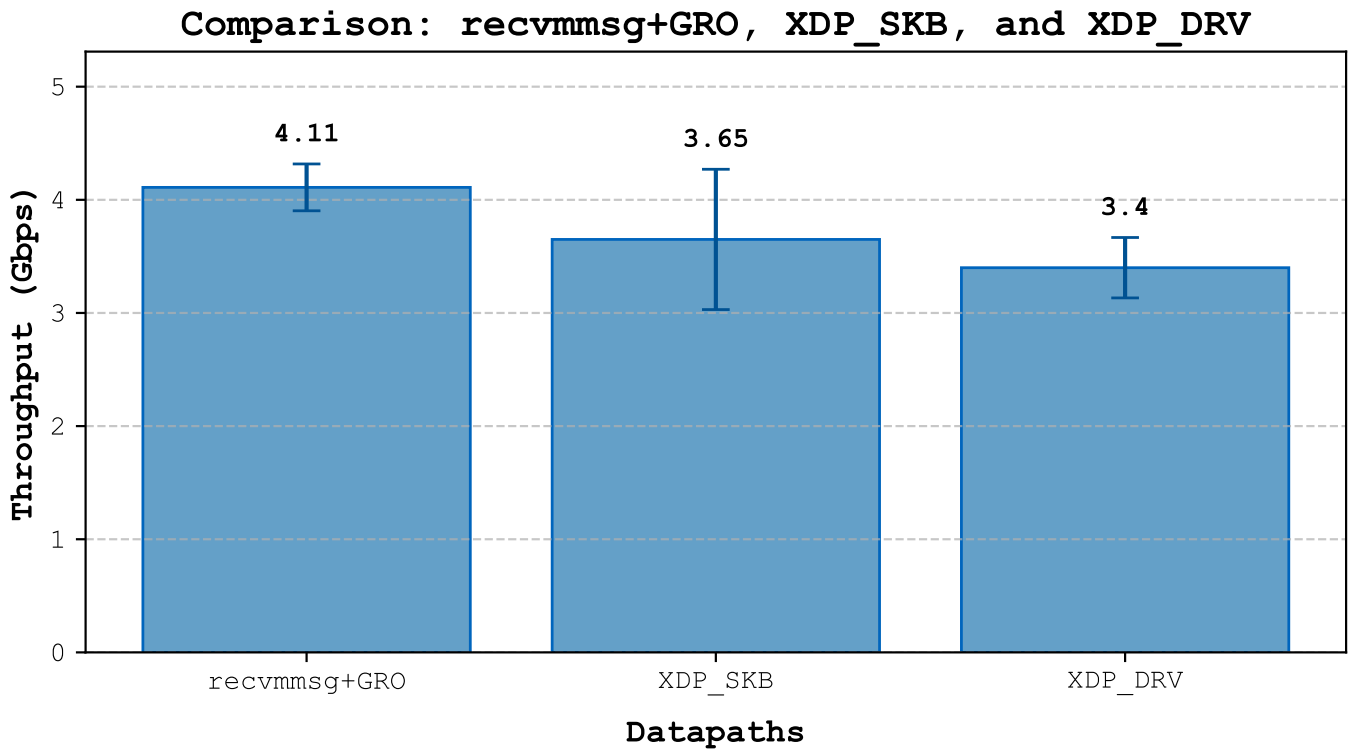
Measurements

- Start with baseline MsQuic datapath (recvmmsg+GRO)
- Switch datapath to AF_XDP XDP_SKB mode



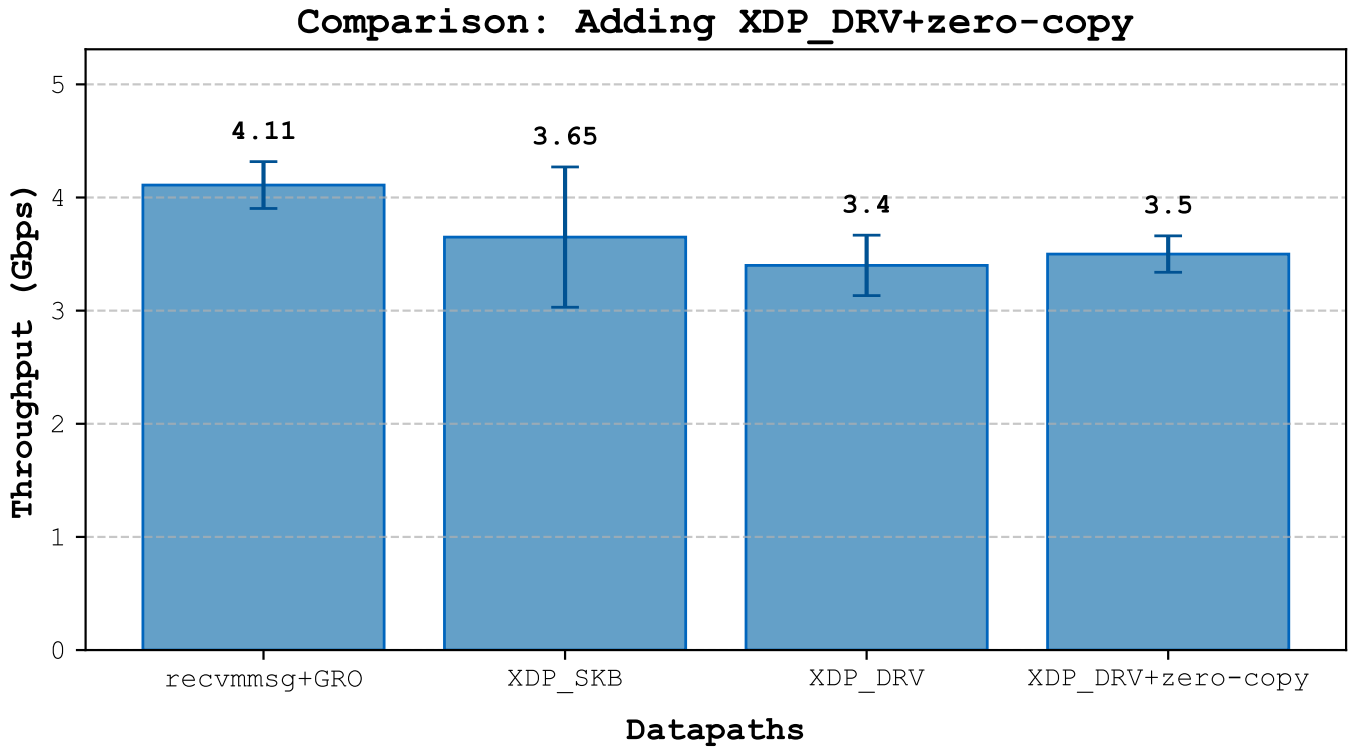
Measurements

- Move the XDP hook point closer to the driver with XDP_DRV
- No longer constructing sk_buffs, so ideally less overhead per packet



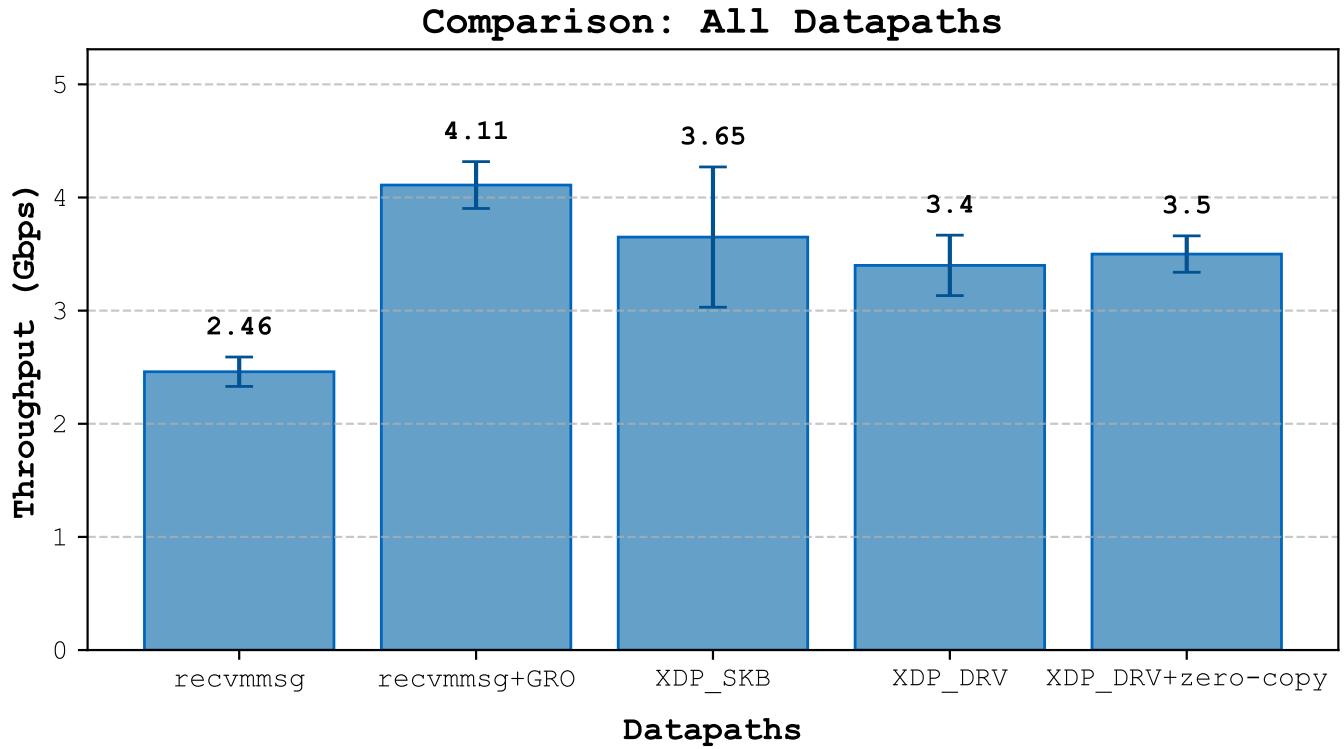
Measurements

- Eliminate an unnecessary additional copy to userspace with XDP_ZEROCOPY
- NIC now DMAs packet directly into UMEM frame



Measurements

- GRO is the main reason the default datapath is so fast
- Can we use GRO with XDP?



Related Work

- DPDK[5] enables kernel bypass and allows application to connect directly to the NIC, reducing kernel overhead
- lxin-quic[6] brings QUIC into the kernel, reducing context switches and copying between kernel and userspace
- *QUIC Performance Improvements Using Socket Optimizations and Kernel Bypass*[7] investigates methods to improve QUIC performance by means of patching MsQuic with DPDK and using socket optimizations

[5] https://www.dpdk.org/

[6] https://github.com/lxin/quic

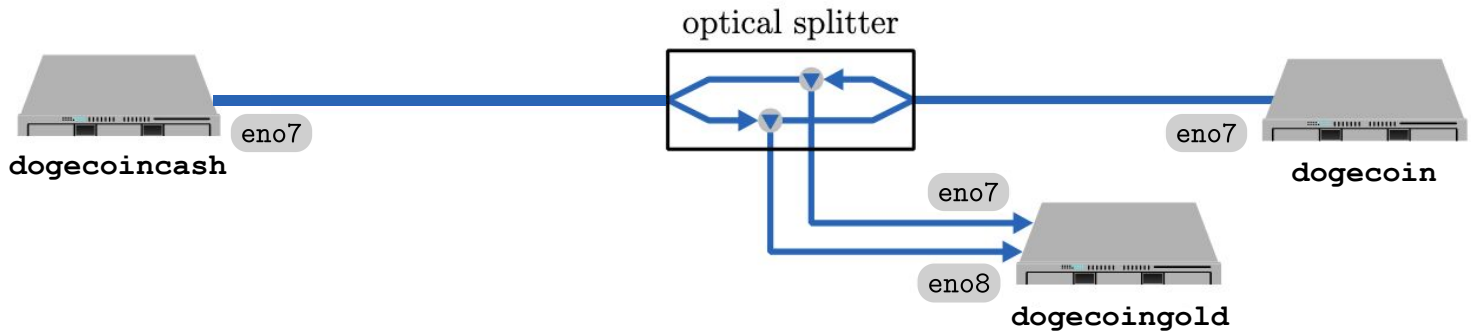
[7] Späth, J. QUIC performance improvements using socket optimizations and kernel bypass. Master's thesis, TU Munich, 2023.

Conclusion

- AF_XDP is not able to achieve a higher throughput over recvmmsg()+GRO
- XDP fragmentation support enables AF_XDP+GRO to potentially further increase performance
- AF_XDP represents one of many methods to reduce packet IO, each with its strengths and weaknesses

Next Steps

- Investigate AF_XDP latency with clock-sychronized splitter and client to measure packet latency based on timing of arrived packets
- Create taxonomy for QUIC optimizations in terms of implementation complexity, performance improvement potential, unwanted side effects, flexibility, privilege level usage, and dynamic vs. static loading



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